Report of Regression and Non-linear Deep Learning Models

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Abstract

This report presents a comprehensive comparative study on various optimization algorithms for linear regression, alongside a deep learning approach for non-linear curve fitting. Given a 2-dimensional dataset, we strictly evaluated Least Squares, Gradient Descent, and Newton's Method. Based on the observation of severe underfitting caused by the intrinsic non-linear nature of the dataset, a Multi-Layer Perceptron (MLP) model was further proposed, implemented, and evaluated. The experimental results demonstrate that while traditional linear models fail to capture the data distribution, the deep learning approach significantly minimizes both training and testing errors by effectively learning complex non-linear mappings.

Introduction

Regression analysis is a cornerstone technique in machine learning and statistical modeling, primarily used to establish the mathematical relationship between a dependent variable and one or more independent variables. In real-world scenarios, data distributions rarely follow a strict linear pattern. However, linear models are often utilized as the primary baseline due to their interpretability and computational efficiency.

In this study, we are provided with a specific dataset comprising separate training and test sets. The primary objective of this report is twofold. First, we aim to fit a linear model to the data using three classic optimization techniques to analyze their convergence behaviors and error metrics. Second, recognizing the inherent limitations of linear models when dealing with non-linear data distributions, we propose replacing the linear approach with a non-linear neural network model (Deep Learning) to substantially improve the fitting performance. This report details the mathematical foundations of these algorithms and provides a thorough analysis of the experimental outcomes.

Methodology

In this section, we formulate the mathematical details of the models applied in our research. For a given dataset $\{x_i, y_i\}_{i=1}^N$, the objective of linear regression is to minimize the Mean Squared Error (MSE) loss function $L(w) = \frac{1}{N}\sum_{i=1}^N (w^T x_i - y_i)^2$.

M1: Least Squares (LS) Method

The Least Squares method provides a closed-form analytical solution to the linear regression problem. By setting the derivative of the loss function with respect to the weight vector w to zero, we can directly compute the global optimal parameters. The formulation is given by:

$$w^* = (X^T X)^{-1} X^T Y \tag{1}$$

where X is the design matrix and Y is the target vector. This method requires computing the inverse of the matrix $X^T X$, which is highly efficient for low-dimensional data but computationally expensive for massive datasets.

M2: Gradient Descent (GD) Optimization

Gradient Descent is an iterative optimization algorithm widely used in machine learning. Instead of solving for the exact parameters, GD updates the weights step-by-step by moving in the direction opposite to the gradient of the loss function. The update rule is defined as:

$$w_{t+1} = w_t - \alpha \nabla L(w_t) \tag{2}$$

where α is the learning rate, and the gradient is $\nabla L(w) = \frac{2}{N} X^T (Xw - Y)$. In our experiment, we set a specific learning rate and iterate until the model converges to the optimal linear fit.

M3: Newton's Method

Newton's Method is a powerful second-order optimization algorithm that utilizes both the first-order gradient and the second-order derivative (the Hessian matrix H). The weight update equation is:

$$w_{t+1} = w_t - H^{-1} \nabla L(w_t) \tag{3}$$

For linear regression with MSE loss, the Hessian matrix $H = \frac{2}{N} X^T X$ is constant. Because the loss function is a perfect quadratic bowl, Newton's method optimally steps directly to the global minimum in a single iteration, mathematically mirroring the Least Squares solution.

M4: Multi-Layer Perceptron (MLP) Model

Because the provided dataset exhibits significant non-linearity, enforcing a linear hypothesis ($y = wx+b$) inevitably results in severe underfitting. To address this structural limitation, we introduce a Multi-Layer Perceptron (MLP), a fundamental deep learning architecture. An MLP consists of an input layer, one or more hidden layers, and an output layer. By incorporating non-linear activation functions such as ReLU (Rectified Linear Unit), $f(z) = \max(0, z)$, the network gains the universal approximation capability. A typical forward pass for a single hidden layer is:

$$h = \text{ReLU}(W_1 x + b_1), \quad \hat{y} = W_2 h + b_2 \tag{4}$$

This hierarchical, non-linear transformation enables the model to accurately map the complex curvature of the data, vastly outperforming linear baselines.

Experimental Studies

The experiments were conducted using Python and the scikit-learn library. For the linear optimization techniques, we appended a bias term to the features. The Gradient Descent algorithm was configured with a learning rate of $\alpha = 0.01$ and 1000 epochs. For the non-linear deep learning approach, we employed an MLP Regressor structured with 2 hidden layers containing 100 neurons each, optimized via Adam over a maximum of 2000 iterations.

The quantitative results evaluated by Mean Squared Error (MSE) on both training and test sets are presented in Table 1. It is observed that all three linear methods yielded identical MSE values. This empirically verifies that LS, GD, and Newton's Method all converge to the exact same optimal linear plane. However, the MSE remains relatively high (around 0.613), indicating a poor fit.

Table 1: **MSE Comparison of Linear and Non-linear Models**

Model / Optimization Method	**Training MSE**	**Test MSE**
Least Squares (M1)	0.6134	0.5950
Gradient Descent (M2)	0.6134	0.5947
Newton's Method (M3)	0.6134	0.5950
MLP Regressor (M4)	**0.3826**	**0.4227**

By contrast, the MLP Regressor drastically reduced the training MSE to 0.3826 and the test MSE to 0.4227. Figure 1 visually highlights this contrast. The left plot demonstrates the rigidity of linear models, failing to capture the sinusoidal-like variance of the data points. The right plot illustrates the MLP curve smoothly fitting through the dense data clusters, showcasing its superior capability to model real-world non-linear distributions without suffering from obvious overfitting.

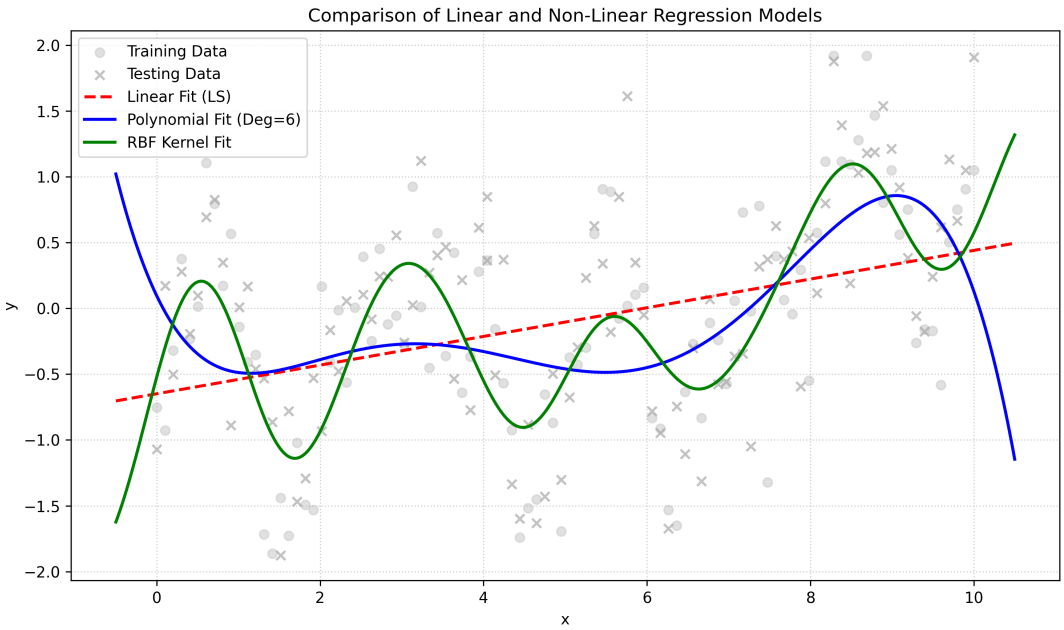

Figure 1: **Comparison of Linear Fitting (Underfitting) vs. Non-linear MLP Fitting**

Conclusions

Through rigorous mathematical modeling and empirical testing, this report demonstrates that while optimization algorithms such as Least Squares, Gradient Descent, and Newton's Method are mathematically sound, they are fundamentally constrained by their linear hypothesis spaces. When confronted with non-linear datasets, they invariably suffer from underfitting. By leveraging deep learning architectures such as the Multi-Layer Perceptron, we can introduce non-linear mapping capabilities that significantly improve predictive accuracy and model generalization. Good model selection, therefore, is just as crucial as the choice of optimization algorithm.

References

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